

Strassen Reformulation

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1 Intro

The purpose of this project was to investigate whether it would be possible to find a way to multiply two 2×2 matrices with only 6 multiplication operations. In 1969, Volker Strassen discovered that it is possible to perform this operation only using 7 multiplications, and so I began my journey there. On the night of Mon, Oct 6, 2025 around 2 AM, after several attempts, I came upon V1. I constructed a new output matrix C' by permuting the entries of C . From this new matrix, I inferred corresponding matrices A' and B' , and then applied Strassen's construction.. I then wrote down the new p' , and I found that although results were in the wrong locations, it was possible to reconstruct C using the p' .

Soon after this I came upon the Alternate Multiplication, as I decided to calculate C , using the $C_{(i,i)}$ that required 2 terms. This lead to 8-multiplication formulation being found.

On July 5, 2026, I came back to this problem, as I had an intuition that there were two more possible variations of Strassen's algorithm. By using the same trick as before, I was able to build the alternate C matrices, and their respective A , and B matrices.

As of currently ChatGPT, says "Subsequent investigation suggests that these reformulations are equivalent to known symmetries of Strassen's algorithm rather than fundamentally new matrix multiplication algorithms".

Additionally, using a symbolic verification program, all 24 permutations of the output matrix were examined. Only four correspond directly to valid matrix products under the reconstruction method used in this paper.

Not every permutation of C corresponds to a valid matrix product. The diagonal entries of C' determine the rows of A' and the columns of B' . Consequently, the diagonal entries must use different rows of A and different columns of B . This constraint explains why only four of the twenty-four possible permutations produce valid matrix products.

2 Standard Multiplication (8 Mults)

Let $\mathbf{A} =$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (1)$$

Let $\mathbf{B} =$

$$\begin{bmatrix} e & f \\ g & h \end{bmatrix} \quad (2)$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix} \quad (3)$$

3 Strassen Multiplication (7 Mults)

Terms:

$$p_1 = a(f - h)$$

$$p_2 = (a + b)h$$

$$p_3 = (c + d)e$$

$$p_4 = d(g - e)$$

$$p_5 = (a + d)(e + h)$$

$$p_6 = (b - d)(g + h)$$

$$p_7 = (a - c)(e + f)$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} p_5 + p_4 - p_2 + p_6 & p_1 + p_2 \\ p_3 + p_4 & p_1 + p_5 - p_3 - p_7 \end{bmatrix} \quad (4)$$

Work:

$$C_{(1,1)} =$$

$$\begin{aligned} & p_5 + p_4 - p_2 + p_6 = \\ & (a + d)(e + h) + d(g - e) - (a + b)h + (b - d)(g + h) = \\ & ae + ah + de + dh + dg - de - ah - bh + bg + bh - dg - dh = \\ & ae + \cancel{ah} + \cancel{de} + \cancel{dh} + \cancel{dg} - \cancel{de} - \cancel{ah} - \cancel{bh} + bg + \cancel{bh} - \cancel{dg} - \cancel{dh} = \\ & ae + bg \end{aligned}$$

$$C_{(1,2)} =$$

$$\begin{aligned} & p_1 + p_2 = \\ & a(f - h) + (a + b)h = \\ & af - ah + ah + bh = \\ & af - \cancel{ah} + \cancel{ah} + bh = \\ & af + bh \end{aligned}$$

$$C_{(2,1)} =$$

$$\begin{aligned} & p_3 + p_4 = \\ & (c + d)e + d(g - e) = \\ & ce + de + dg - de = \\ & ce + \cancel{de} + dg - \cancel{de} = \\ & ce + dg \end{aligned}$$

$$C_{(2,2)} =$$

$$\begin{aligned} & p_1 + p_5 - p_3 - p_7 = \\ & a(f - h) + (a + d)(e + h) - (c + d)e - (a - c)(e + f) = \\ & af - ah + ae + ah + de + dh - ce - de - ae - af + ce + cf \\ & \cancel{af} - \cancel{ah} + \cancel{ae} + \cancel{ah} + \cancel{de} + dh - \cancel{ce} - \cancel{de} - \cancel{ae} - \cancel{af} + \cancel{ce} + cf = \\ & cf + dh \end{aligned}$$

4 V1 Derivation

The technique used in this project begins by rearranging the entries of the product matrix

$$C = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}.$$

For example, one may form a new matrix C' by permuting the four entries of C , ensuring that $ae + bg$ and $cf + dh$ stay diagonal. After choosing a candidate C' , the goal is to build A', B' , such that

$$C' = A'B'.$$

Ex.

$$C' = \begin{bmatrix} ce + dg & cf + dh \\ ae + bg & af + bh \end{bmatrix} \quad (5)$$

Thus

$$A' = \begin{bmatrix} c & d \\ a & b \end{bmatrix} \quad (6)$$

and

$$B' = \begin{bmatrix} e & f \\ g & h \end{bmatrix} \quad (7)$$

Do remember that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (8)$$

Table 1: Comparison of Strassen's original intermediate products and the corresponding products after applying the V1 transformation. These equations can be found quickly by just locating and swapping the terms of p based on the new matrix.

Product	Original Strassen	V1 Reformulation
p_1	$a(f - h)$	$c(f - h)$
p_2	$(a + b)h$	$(c + d)h$
p_3	$(c + d)e$	$(a + b)e$
p_4	$d(g - e)$	$b(g - e)$
p_5	$(a + d)(e + h)$	$(b + c)(e + h)$
p_6	$(b - d)(g + h)$	$(d - b)(g + h)$
p_7	$(a - c)(e + f)$	$(c - a)(e + f)$

However, if we were to use these terms as intended in Strassen's Algorithm we would end up forming C' , thus we must change the location of the p terms to correctly calculate C .

5 Strassen V1 Multiplication (7 Mults)

Terms:

$$a_1 = c(f - h)$$

$$a_2 = (c + d)h$$

$$a_3 = (a + b)e$$

$$a_4 = b(g - e)$$

$$a_5 = (c + b)(e + h)$$

$$a_6 = (d - b)(g + h)$$

$$a_7 = (c - a)(e + f)$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} a_3 + a_4 & a_1 + a_5 - a_3 - a_7 \\ a_5 + a_4 - a_2 + a_6 & a_1 + a_2 \end{bmatrix} \quad (9)$$

$$C_{(1,1)} =$$

$$a_3 + a_4 =$$

$$(a + b)e + b(g - e) =$$

$$ae + be + bg - be =$$

$$ae + \cancel{be} + bg - \cancel{be} =$$

$$ae + bg$$

$$C_{(1,2)} =$$

$$a_1 + a_5 - a_3 - a_7 =$$

$$c(f - h) + (c + b)(e + h) - (a + b)e - (c - a)(e + f) =$$

$$cf - ch + ce + ch + be + bh - ae - be - ce - cf + ae + af =$$

$$\cancel{cf} - \cancel{ch} + \cancel{ce} + \cancel{ch} + \cancel{be} + bh - \cancel{ae} - \cancel{be} - \cancel{ce} - \cancel{cf} + \cancel{ae} + af =$$

$$af + bh$$

$$C_{(2,1)} =$$

$$a_5 + a_4 - a_2 + a_6 =$$

$$(c + b)(e + h) + b(g - e) - (c + d)h + (d - b)(g + h) =$$

$$ce + ch + be + bh + bg - be - ch - dh + dg + dh - bg - bh =$$

$$ce + \cancel{ch} + \cancel{be} + \cancel{bh} + \cancel{bg} - \cancel{be} - \cancel{ch} - \cancel{dh} + dg + \cancel{dh} - \cancel{bg} - \cancel{bh} =$$

$$ce + dg$$

$$C_{(2,2)} =$$

$$a_1 + a_2 =$$

$$c(f - h) + (c + d)h =$$

$$cf - ch + ch + dh =$$

$$\cancel{cf} - \cancel{ch} + \cancel{ch} + dh =$$

$$cf + dh$$

6 Strassen V2 Multiplication (7 Mults)

Terms:

$$r_1 = a(e - g)$$

$$r_2 = (a + b)g$$

$$r_3 = (c + d)f$$

$$r_4 = d(h - f)$$

$$r_5 = (a + d)(f + g)$$

$$r_6 = (b - d)(h + g)$$

$$r_7 = (a - c)(f + e)$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} r_1 + r_2 & r_5 + r_4 - r_2 + r_6 \\ r_1 + r_5 - r_3 - r_7 & r_3 + r_4 \end{bmatrix} \quad (10)$$

$$C_{(1,1)} =$$

$$r_1 + r_2 =$$

$$ae + bg$$

$$C_{(1,2)} =$$

$$r_5 + r_4 - r_2 + r_6 =$$

$$af + bh =$$

$$C_{(2,1)} =$$

$$r_1 + r_5 - r_3 - r_7 =$$

$$ce + dg =$$

$$C_{(2,2)} =$$

$$r_3 + r_4 =$$

$$dh + cf = cf + dh$$

7 Strassen V3 Multiplication (7 Mults)

Terms:

$$e_1 = c(e - g)$$

$$e_2 = (c + d)g$$

$$e_3 = (a + b)f$$

$$e_4 = b(h - f)$$

$$e_5 = (c + b)(f + g)$$

$$e_6 = (d - b)(h + g)$$

$$e_7 = (c - a)(f + e)$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} e_1 + e_5 - e_3 - e_7 & e_3 + e_4 \\ e_1 + e_2 & e_5 + e_4 - e_2 + e_6 \end{bmatrix} \quad (11)$$

$$C_{(1,1)} =$$

$$e_1 + e_5 - e_3 - e_7 =$$

$$ae + bg$$

$$C_{(1,2)} =$$

$$e_3 + e_4 =$$

$$af + bh =$$

$$C_{(2,1)} =$$

$$e_1 + e_2 =$$

$$ce + dg =$$

$$C_{(2,2)} =$$

$$e_5 + e_4 - e_2 + e_6 =$$

$$cf + dh$$

8 Alternate Multiplication (8 Mults)

Terms:

$$w_1 = (a + b)e$$

$$w_2 = b(g - e)$$

$$w_3 = a(f - h)$$

$$w_4 = (a + b)h$$

$$w_5 = (c + d)e$$

$$w_6 = d(g - e)$$

$$w_7 = c(f - h)$$

$$w_8 = (c + d)h$$

Let $\mathbf{C} = \mathbf{AB} =$

$$\begin{bmatrix} w_1 + w_2 & w_3 + w_4 \\ w_5 + w_6 & w_7 + w_8 \end{bmatrix} \quad (12)$$

$$C_{(1,1)} =$$

$$\begin{aligned} w_1 + w_2 &= \\ (a + b)e + b(g - e) &= \\ ae + bebg - be &= \\ ae + \cancel{be} + bg - \cancel{be} &= \\ ae + bg & \end{aligned}$$

$$C_{(1,2)} =$$

$$\begin{aligned} w_3 + w_4 &= \\ a(f - h) + (a + b)h &= \\ af - ah + ah + bh &= \\ af - \cancel{ah} + \cancel{ah} + bh &= \\ af + bh & \end{aligned}$$

$$C_{(2,1)} =$$

$$\begin{aligned} w_5 + w_6 &= \\ (c + d)e + d(g - e) &= \\ ce + de + dg - de &= \\ ce + \cancel{de} + dg - \cancel{de} &= \\ ce + dg & \end{aligned}$$

$$C_{(2,2)} =$$

$$\begin{aligned} w_7 + w_8 &= \\ c(f - h) + (c + d)h &= \\ cf - ch + ch + dh &= \\ cf - \cancel{ch} + \cancel{ch} + dh &= \\ cf + dh & \end{aligned}$$